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MAGNETIC MEMORY USING SPIN CURRENT, OPERATING METHOD THEREOF, AND ELECTRONIC APPARATUS INCLUDING MAGNETIC MEMORY

Abstract

Provided are a magnetic memory using a spin current, an operating method thereof, and/or an electronic apparatus including the magnetic memory. The magnetic memory includes, first and second wirings spaced apart from each other and intersecting each other, and a data storage layer between the first and second wirings. The data storage layer includes a pinned layer with a fixed magnetic moment, a free layer spaced apart from the pinned layer and not having a fixed magnetic moment, and an insulating tunnel barrier layer provided between the pinned layer and the free layer. Among the first and second wirings, the wiring contacting the free layer includes a conductive wiring having no spin Hall effect, and the free layer includes a two-dimensional material which at room temperature has a spin Hall effect, magnetic properties, and metal properties. The two-dimensional material includes a two-dimensional van der Waals material.

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Background/Summary

CROSS-REFERENCE TO RELATED APPLICATION [0001] This application is a continuation of U.S. patent application Ser. No. 17/964,373, filed on Oct. 12, 2022, which is based on and claims priority under 35 U.S.C. § 119 to Korean Patent Application No. 10-2022-0053811, filed on Apr. 29, 2022, in the Korean Intellectual Property Office, the disclosure of each of which is incorporated by references herein in their entirety.

BACKGROUND

[0002] Various example embodiments relate to a memory device, and more particularly, to a magnetic memory using a spin current, an operating method thereof, and/or an electronic apparatus including the magnetic memory.

[0003] A magnetic memory device records or reads information using a tunneling magnetoresistance (TMR) phenomenon.

[0004] When the magnetic moment directions of the two magnetic layers included in a magnetic tunnel junction (MTJ) of a magnetic memory device are parallel to each other, the magnetic memory device has low resistance, and when the magnetic moment directions are parallel opposite to each other, the magnetic memory device has higher resistance.

[0005] An MTJ includes two magnetic layers, such as a pinned layer, which is also referred to as a fixed layer and may have a fixed direction of the magnetic moment, and a free layer in which the direction of the magnetic moment may be switched.

[0006] A magnetic memory device using a spin transfer torque for a writing operation for determining a direction of the free layer is referred to as a spin transfer torque-magnetic random access memory (STT-MRAM), and a magnetic memory device using a spin orbit torque is referred to as a spin orbit torque magnetic random-access memory (SOT-MRAM).

[0007] The STT-MRAM is applied to an embedded memory, and performs a read operation and a write operation using a current flowing through the MTJ.

[0008] The SOT-MRAM has three terminal structures for switching the free layer with a current flowing in a spin orbit torque (SOT) layer formed below the MTJ.

[0009] The SOT-MRAM compared to the STT-MRAM enables fast operation of 1 nanosecond or less (<1 nsec), and the write path and the read path are separated so that a write bias is not directly applied to the tunnel barrier layer of the MTJ, and thus durability may be improved.

SUMMARY

[0010] Provided is a magnetic memory device using a spin current, having high-speed switching characteristics and high durability.

[0011] Alternatively or additionally, provided is a magnetic memory device using a spin current, capable of increasing an integration degree.

[0012] Alternatively or additionally, provided is a magnetic memory device using a spin current, capable of lowering an operating current density.

[0013] Alternatively or additionally, provided are a method of manufacturing a magnetic memory device and an operating method of the magnetic memory device.

[0014] Alternatively or additionally, provided is an electronic apparatus including a magnetic memory device.

[0015] Additional aspects will be set forth in part in the description which follows and, in part, will be apparent from the description, or may be learned by practice of the presented embodiments of the disclosure.

[0016] According to some example embodiments, a magnetic memory using a spin current includes a first wiring and a second wiring that are spaced apart from each other, and a data storage layer between the first and second wirings. The data storage layer includes a pinned layer having a fixed magnetic moment and a free layer spaced apart from the pinned layer and having no fixed magnetic moment. Among the first and second wirings, the wiring contacting the free layer is or includes a conductive wiring having no spin Hall effect, and the free layer includes a two-dimensional material having a spin Hall effect, magnetic properties, and metal properties. In some examples, the two-dimensional material may include a two-dimensional van der Waals material. In some examples, the two-dimensional van der Waals material may include one or more of Fe.sub.

(x)GeTe.sub.2 (x is 3 to 5), CrTe.sub.2, and VSe.sub.2. In some examples, each of the first and second wirings may be provided with a plurality of wirings, and the data storage layer may be provided for each cross area of the plurality of first wirings and the plurality of second wirings. In some examples, the data storage layer may include an MTJ layer in each cross area and may further include an insulating tunnel barrier layer provided between the pinned layer and the free layer. In some examples, the tunnel barrier layer may include one or more of an MgO layer, a boron nitride (BN) layer, or an aluminum oxide layer. In some examples, a material of the pinned layer may be the same as or different from the material of the free layer.

[0017] According to some example embodiments, there is provided an operating method of a spin orbit torque-magnetic random access memory (SOT-MRAM), wherein the method includes applying a first operating current to a wiring contacting the free layer of the first and second wirings, a first spin current being generated to the free layer by the first operating current; and applying a second operating current to a wiring not contacting the free layer of the first and second wirings, a second spin current being generated to the free layer by the second operating current. In some examples, the second operating current may be less than the first operating current. In some examples, the first and second operating currents may be or may correspond to either or both of a write current or an erasure current.

[0018] According to some example embodiments, there is provided an operating method of a magnetic memory including a magnetic tunnel junction (MTJ) including a pinned layer and a free layer, the method comprising: lowering a coercive force of the free layer by using a first spin current; and switching a magnetic moment of the free layer in which the coercive force is lowered by using a second spin current. In some examples, the first and second spin currents may be or may correspond to currents for data recording and/or current for data erasure.

[0019] In some examples, the process of lowering the coercive force of the free layer may include applying the first operating current to the wiring contacting the free layer.

[0020] In some examples, the process of switching the magnetic moment of the free layer includes a process of applying the second operating current to the wiring contacting the pinned layer. In

some examples, the free layer may include a two-dimensional van der Waals material having a spin Hall effect at room temperature and magnetic properties and metallic properties. In some examples, the wiring contacting the free layer may include a metal wiring having no spin Hall effect.

[0021] According to some example embodiments, there is provided an electronic apparatus including a memory, wherein the memory includes the magnetic memory according to various example embodiments described above.

[0022] According to some example embodiments, a magnetic memory includes a first wiring extending in a first direction, a second wiring extending in a second direction over the first wiring, and a data storage layer at an intersection between the first wiring and the second wiring and contacting the first and second wirings. The data storage layer comprises a pinned layer having a fixed magnetic moment, a free layer apart from the pinned layer and configured to rotate the magnetic moment, and an insulating tunnel barrier layer between the pinned layer and the free layer. One of the first and second wirings contacts the free layer and includes a conductive wiring that does not have a spin Hall effect, and the free layer comprises a two-dimensional material having a spin Hall effect, magnetic properties, and metallic properties. The first direction may be orthogonal with the second direction. The first wiring may correspond to a bit line of the magnetic memory, and the second wiring corresponds to a word line of the magnetic memory. A spin orbit torque-magnetic random access memory (SOT-MRAM) may comprise a plurality of the magnetic memories, wherein the plurality of magnetic memories are arranged in a cross-point lattice.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

[0023] The above and other aspects, features, and advantages of certain example embodiments of the disclosure will be more apparent from the following description taken in conjunction with the accompanying drawings, in which:

[0024] FIG. 1 is a cross-sectional view illustrating a unit cell of a magnetic memory device using a spin current, according to some example embodiments, which is a unit cell of a spin orbit torque magnetic random-access memory (SOT-MRAM);

[0025] FIG. 2 is a plan view of a unit cell of the SOT-MRAM of FIG. 1;

[0026] FIG. 3 shows an MTJ cell array of a SOT-MRAM in which the MTJ cell array consists of the unit cells of the SOT-MRAM of FIG. 1;

[0027] FIGS. 4A to 4D are graphs illustrating simulation results for switching characteristics of a 2D van der Waals material used as a free layer of an MTJ cell of a SOT-MRAM according to some example embodiments;

[0028] FIGS. 5 to 7 are perspective views illustrating stepwise operation processes of the SOT-MRAM according to some example embodiments;

[0029] FIG. 8 is a block diagram illustrating a first electronic apparatus according to some example embodiments;

[0030] FIG. 9 is a block diagram illustrating a second electronic apparatus according to some example embodiments;

[0031] FIG. 10 is a block diagram illustrating a third electronic apparatus according to some example embodiments; and

[0032] FIG. 11 is a block diagram schematically illustrating a fourth electronic apparatus according to some example embodiments.

DETAILED DESCRIPTION

[0033] Reference will now be made in detail to certain example embodiments, examples of which are illustrated in the accompanying drawings, wherein like reference numerals refer to like elements throughout. In this regard, the example embodiments may have different forms and

should not be construed as being limited to the descriptions set forth herein. Accordingly, the embodiments are merely described below, by referring to the figures, to explain aspects. As used herein, the term “and/or” includes any and all combinations of one or more of the associated listed items. Expressions such as “at least one of,” when preceding a list of elements, modify the entire list of elements and do not modify the individual elements of the list.

[0034] Although numerical terms such as “first,” “second,” “third,” etc., may be used herein to describe various elements, components, regions, layers, and/or sections, these numerical terms are only used to distinguish one element, component, region, layer, or section, from another region, layer, or section; and the elements, components, regions, layers, and/or sections, and should not be otherwise limited by these numerical terms. Thus, a first element, component, region, layer, or section, discussed below may be termed a second element, component, region, layer, or section, without departing from the scope of this disclosure.

[0035] When the terms “about” or “substantially” are used in this specification in connection with a numerical value, it is intended that the associated numerical value includes a manufacturing tolerance (e.g., $\pm 10\%$) around the stated numerical value. Moreover, when the words “generally” and “substantially” are used in connection with geometric shapes, it is intended that precision of the geometric shape is not required but that latitude for the shape is within the scope of the disclosure. Further, regardless of whether numerical values or shapes are modified as “about” or “substantially,” it will be understood that these values and shapes should be construed as including a manufacturing or operational tolerance (e.g., $\pm 10\%$) around stated numerical values or shapes.

[0036] Hereinafter, a magnetic memory using a spin current according to an example embodiment, an operating method thereof, and an electronic apparatus including the magnetic memory device will be described in detail with reference to the accompanying drawings. In the following description, the thickness of the layers or regions shown in the drawings may be somewhat exaggerated for clarity of the specification. In addition, various embodiments described below are merely examples and various modifications are possible from embodiments. Further, in the layer structure described below, the expressions referred to as “upper portion” or “on” may be referred to as being in contact directly, as well as being one above another in non-contact. In the following description, like reference numerals refer to like elements in each drawing.

[0037] First, a magnetic memory using a spin current according to some example embodiments will be described.

[0038] FIG. 1 is a cross-sectional view illustrating a magnetic memory device using a spin current, according to some example embodiments, which is a unit cell of a spin orbit torque magnetic random access memory (SOT-MRAM), and FIG. 2 is a plan view thereof. FIG. 1 is a cross-sectional view taken along the line 1-1' of FIG. 2.

[0039] Referring to FIGS. 1 and 2, a unit cell **100** of the SOT-MRAM may include a first wiring **120**, a second wiring **170**, and a data storage layer MS. In some examples, the unit cell **100** may be formed to further include other members or components. The first wiring **120** and the second wiring **170** are spaced apart from each other. The first wiring **120** and the second wiring **170** may be arranged to cross each other (e.g. to cross each other in a perpendicular manner), and the first and second wirings **120** and **170** may cross each other vertically. One of the first and second wirings **120** and **170** may be or may correspond to a bit line or include a bit line, and the other may be or may correspond to a word line or include a word line. In some examples, the first wiring **120** may be a bit line, e.g. a column and the second wiring **170** may be a word line, e.g. a row. In some examples, the first wiring **120** may be a word line and the second wiring **170** may be a bit line. The first and/or second wirings **120** and **170** may be or include conductive wirings used in a semiconductor device and/or an electronic apparatus, for example, may be metal wirings having no spin Hall effect or substantially no spin Hall effect or include such metal wirings. In some examples, the first and/or second wirings **120** and **170** may be copper (Cu) wirings or may include copper wirings, but are not limited to the copper wirings.

[0040] The data storage layer MS is formed or arranged between the first wiring **120** and the second wiring **170**. The first surface **S1** of the data storage layer MS is connected to (or directly connected to) one of the first and second wirings **120** and **170**, and the second surface **S2** is connected to (or directly connected to) the other of the first and second wirings **120** and **170**. Here, the term “connection” may include a contact or a contact state. In some examples, the first surface **S1** may be the bottom surface of the data storage layer MS and may be in direct contact with the first wiring **120**. In some examples, the second surface **S2** may be the top surface of the data storage layer MS and may be in direct contact with the first wiring **170**. The “bottom surface” and the “top surface” may be referred to differently depending on the viewpoint of viewing the unit cell **100**. For example, depending on the viewpoint, the “bottom surface” may be an upper surface or a side surface, and the “top surface” may be a bottom surface or a side surface.

[0041] The first and second surfaces **S1** and **S2** may be spaced apart from each other in a direction perpendicular to or substantially perpendicular to the first wiring **120** and/or second wiring **170**, and may face each other. Alternatively or additionally, the first and second surfaces **S1** and **S2** may be parallel or substantially parallel to each other, but are not limited thereto. Alternatively or additionally, the first surface **S1** and/or second surface **S2** may be parallel or substantially parallel to the first wiring **120** and/or second wiring **170**.

[0042] The entire data storage layer MS may be provided on the first wiring **120**. For example, the entire data storage layer MS may be arranged on the top surface of the first wiring **120**, but is not limited thereto.

[0043] In some examples, the data storage layer MS has a first width **W1** in a direction (e.g. a Y-axis direction) perpendicular to the longitudinal direction (e.g. an X-axis direction) of the first wiring **120**, and may have a second width **W2** in the longitudinal direction of the first wiring **120**. The first and second widths **W1** and **W2** may be the same as or different from each other. The first wiring **120** may have a third width **W3** in a direction perpendicular to the longitudinal direction. The first width **W1** of the data storage layer MS and the third width **W3** of the first wiring **120** may be the same or may be different from each other. In some examples, the first width **W1** of the data storage layer MS may be less than the third width **W3** of the first wiring **120**.

[0044] In some examples, the data storage layer MS may be or may include or correspond to a magnetic tunnel junction (MTJ) layer.

[0045] The data storage layer MS may include a first layer **130**, a second layer **140**, and a third layer **150** sequentially stacked on the first wiring **120** in the direction of the second wiring **170**. In some examples, the sizes of the first to third layers **130**, **140**, and **150** may be the same. The thicknesses of the first to third layers **130**, **140**, and **150** may be different from each other, but some of the layers may be the same. In some examples, the first layer **130** and the third layer **150** may have the same thickness.

[0046] The bottom surface of the first layer **130** directly contacting the first wiring **120** may correspond to the first surface **S1** of the data storage layer MS. The top surface of the third layer **150** directly contacting the second wiring **170** may correspond to the second surface **S2** of the data storage layer MS.

[0047] The first layer **130** and/or the third layer **150** may be a monolayer, or have a layer structure consisting of or having a plurality of monolayers, that is, a multilayer structure. At least some materials of the first and third layers **130** and **150** may be different from the material of the second layer **140**. The first and/or third layers **130** and **150** may include a material layer exhibiting a spin Hall effect, magnetic properties, and metal properties. In some examples, the first and/or third layers **130** and **150** may include a 2D material layer exhibiting ferromagnetic properties and metal properties at room temperature while having a spin Hall effect. As an example of the metal characteristics, the resistance of the 2D material layer may be lower than 500 $\mu\Omega\text{cm}$.

[0048] In some examples, such a 2D material layer may be or may include a 2D van der Waals (hereinbelow, referred to as 2dvW) material layer or may include a 2dvW material layer. In some

examples, the 2dvW material layer may include any one of or at least one of Fe.sub.(x)GeTe.sub.2, CrTe.sub.2, and VSe.sub.2. In some examples, “x” of Fe.sub.(x)GeTe.sub.2 may be 3 to 5, e.g., 3, 4, or 5.

[0049] In some examples, the 2D material layer used as the material of the first layer **130** and/or the third layer **150** may include a material layer that a ratio where the charge current is changed to a spin current, e.g., a spin Hall angle (SHA) is a first value. In some examples, the first value may be greater than 0.3 and less than 1.0. The first value of 0.3 indicates that 30% of the charge current is changed to the spin current.

[0050] In some examples, one of the first layer **130** and the third layer **150** may be or correspond to a pinned layer in which a magnetic moment is fixed or substantially fixed in a first direction, and the other may be or correspond to a free layer in which the magnetic moment may be rotated or switched in the first or second direction by an external action. The magnetic moment may be represented as a magnetic polarization. The second direction may be opposite to (anti-parallel with) the first direction. The first and second directions may be perpendicular to the first and/or second wirings **120** and **170**. In some examples, the action given from the outside may be a magnetic field given from the outside of the data storage layer MS or a current flowing through the other of the first layer **130** and the third layer **150**.

[0051] When the magnetic moment direction of the free layer is aligned in parallel with the magnetic moment direction of the pinned layer, the data storage layer MS becomes a first state having a low electrical resistance. In contrast, when the magnetic moment direction of the free layer is opposite to that of the pinned layer, the data storage layer MS becomes a second state having a high electrical resistance, with a resistance more than that of the low resistance state. One of the first and second states may be considered to be written with data “1” and the other may be considered to be written with data “0”.

[0052] In some examples, the first and/or third layers **130** and **150** include a 2D material layer as described above, and a layer used as the free layer of the first and third layers **130** and **150** may be formed as a monolayer of 2D material. In this case, the operating current (e.g., the write operation current) of the unit cell **100** may be further lowered.

[0053] The second layer **140** is or includes a tunnel barrier layer. In some examples, the second layer **140** may include a material layer having no spin Hall effect or substantially no spin Hall effect. In some examples, the second layer **140** may be an insulating layer or may include such an insulating layer. In some examples, the insulating layer may include, but is not limited to, one or more of a magnesium oxide (e.g., MgO) layer, a boron nitride (BN) layer, or an aluminum oxide layer.

[0054] In FIG. 2, a planar shape of the data storage layer MS is formed to have a rectangular shape, but is not limited thereto. In some examples, the data storage layer MS may be formed to have a circular planar shape or an oval planar shape or a polygonal planar shape.

[0055] In some examples, a separate thin conductive layer may be further provided between one or both of the first and second wirings **120** and **170** and the data storage layer MS. In this case, the first and/or second wirings **120** and **170** may not be in direct contact with the data storage layer MS. The conductive layer may include a metal layer.

[0056] As illustrated in FIG. 1, as an operating voltage is applied to the unit cell **100**, the operating current **110** flows in the first wiring **120**, and the first layer **130** contacting the first wiring **120** has metal characteristics, and thus the operating current **110** also flows in the first layer **130** as indicated by a dotted line **110a**. Accordingly, a spin current may be generated in the first layer **130** by a spin Hall effect. When a separate conductive layer is provided between one or both of the first and second wirings **120** and **170** and the data storage layer MS, the conductive layer may have a thickness that does not interfere or that does not significantly interfere with the operation current **110** flowing through the first layer **130**.

[0057] FIG. 3 illustrates a SOT-MRAM array **300** including unit cells **100** of the SOT-MRAM

shown in FIGS. 1 and 2.

[0058] Referring to FIG. 3, the array 300 includes a plurality of first wirings 120 arranged parallel to each other and a plurality of second wirings 170 arranged parallel to each other. The plurality of first wirings 120 may be arranged to have a dynamically determined, or predetermined distance from each other. The plurality of second wirings 170 may be arranged to have a dynamically determined, or predetermined distance from each other. The first wirings 120 and the second wirings 170 are arranged to vertically intersect each other. A data storage layer MS is provided at each intersection of the first wiring 120 and the second wiring 170, that is, at each crossing area. The array 300 may have a cross-point architecture; however, example embodiments are not limited thereto.

[0059] A plurality of data storage layers MS are arranged on one first wiring 120, and as described above, the first wiring 120 is not formed of a material of a SOT layer having higher electrical resistance than a general metal layer such as a bit line of a conventional SOT-MRAM. In some examples, the second wiring 170 may be arranged in a direction that is not perpendicular to the first wiring 120.

[0060] The first wiring 120 is or includes a metal wiring used as an electrical wiring in a semiconductor device or a general electronic apparatus, and has a much lower electrical resistance than a SOT layer of a conventional SOT-MRAM. In some examples, the operating current density of the SOT-MRAM according to some example embodiments is about 1 MA per square centimeter, which is about 1/10 to 1/100 of existing operating current density of the SOT-MRAM.

[0061] Since the electrical resistance of the first wiring 120 is low, the number of data storage layers MS that may be aligned (or formed) on one first wiring 120 may be greater than of existing SOT-MRAM.

[0062] As a result, in the case of the SOT-MRAM according to some example embodiments, the number of unit cells that can be driven by one driving circuit may be increased compared to the existing SOT-MRAM. Therefore, an area occupied by the driving circuit in the SOT-MRAM according to some example embodiments may be smaller than an area occupied by the driving circuit in the existing SOT-MRAM.

[0063] For example, when the areas occupied by the driving circuits are the same, the degree of integration of the SOT-MRAM according to some example embodiments may be greater than that of the existing SOT-MRAM.

[0064] FIGS. 4A to 4D show experimental results for confirming or determining magnetic properties of a 2dvW material used as a free layer of the data storage layer MS. In the experiments for obtaining the experimental results of FIGS. 4A to 4D, FGT (e.g., Fe.sub.3GeTe.sub.2) was used as the 2dvW material.

[0065] In the experiment, as shown in FIG. 4A, an FGT layer was formed on a silicon (Si) substrate having an upper surface covered with a SiO.sub.2 layer, and then a potential difference was formed between both ends of the formed FGT layer, thereby allowing a current to flow in the FGT layer. At the same time, the resistance of the FGT layer was measured at a point between both ends of the FGT layer in a direction perpendicular to both ends.

[0066] FIG. 4B illustrates a change in the coercive force of the FGT layer according to a current (about 0.05 mA to 2.0 mA) applied to the FGT layer. In FIG. 4B, the horizontal axis represents the strength of the magnetic field and the vertical axis represents the resistance.

[0067] In FIG. 4B, a first graph G1 indicates when the applied current is 0.05 mA or 0.1 mA. The change in the coercive force of the FGT layer when the applied current is 0.05 mA and the change in the coercive force of the FGT layer when the applied current is 0.1 mA are similar to each other and overlap each other on the graph. Accordingly, the changes in the coercive force of the FGT layer when the applied currents are 0.05 mA and 0.1 mA, respectively, are represented by one first graph G1. A second graph to a ninth graph G2-G9 represent respective changes in the coercive force when the applied currents are 0.2 mA, 0.5 mA, 0.75 mA, 1.0 mA, 1.25 mA, 1.5 mA, 1.75 mA

and 2.0 mA.

[0068] FIG. 4B shows that the FGT layer is a ferromagnetic layer having magnetic hysteresis characteristics. In addition, as the magnitude of the applied current increases, the width of a graph representing the magnetic hysteresis characteristic decreases, which suggests that or is consistent with the coercive force H_c of the FGT layer decreases as the magnitude of the current applied to the FGT layer increases. These results show that as the current applied to the FGT layer having a spin Hall effect increases the spin Hall effect of the FGT layer increases (spin current increases), and thus the coercive force of the FGT layer decreases. For example, as the current applied to the FGT layer increases, the spin current in the FGT layer increases, and thus the strength of the magnetic field required for switching the magnetic moment of the FGT layer may decrease. The current applied to the FGT layer may be regarded as a write current in an operation of recording information in the data storage layer MS.

[0069] FIG. 4C illustrates only the separate magnetic hysteresis characteristics of the FGT layer when a large or the maximum current (2.0 mA) and a small or the minimum current (0.05 mA) are applied in FIG. 4B.

[0070] Referring to FIG. 4C, it may be seen that the width of the magnetic hysteresis characteristic graph when a minimum current is applied to the FGT layer is much smaller than the width of the magnetic hysteresis characteristic graph when the maximum current is applied thereto.

[0071] FIG. 4D illustrates that switching occurs between the “1” state and the “0” state even in a small coercive force as a write operation current (I_{write}) applied to the FGT layer increases.

[0072] Next, an operating method of the SOT-MRAM according to some example embodiments will be described with reference to FIGS. 5 to 7. Although FIGS. 5 to 7 illustrate only four unit cells for convenience as parts of the array 300 shown in FIG. 3, an operating method to be described with reference to FIGS. 5 to 7 may be equally applied to the entire area of the array 300. The operating method may be a method of recording information. The information may include bit data “1” or “0”.

[0073] FIG. 5 illustrates a preparation step before information is recorded. In the preparation step, a voltage for information recording is not applied to the data storage layer MS arranged between the first and second wirings 120 and 170.

[0074] For convenience, in the description of FIGS. 5 to 7, the first layer 130 of the data storage layer MS is regarded as a free layer and the third layer 150 is regarded as a pinned layer. In addition, the magnetic hysteresis curves illustrated below the first wiring 120 in FIGS. 5 to 7 show the magnetic hysteresis characteristics of the first layer 130 of the data storage layer MS.

[0075] Referring to FIG. 5, in the preparation stage, there is no current penetrating the second layer 140 of the data storage layer MS, and there is no current flowing in the first layer 130. Accordingly, the magnetic hysteresis characteristic curve of the first layer 130 of each data storage layer MS may have a first width $H1$ in a horizontal direction.

[0076] In this state, as shown in FIG. 6, a first write current I_{w1} is applied to the first wiring 120. The first layer 130 of the data storage layer MS is a 2dvw material layer and has metal characteristics, and the first write current I_{w1} may flow along the first wiring 120 and may flow through the first layer 130 in direct contact with the first wiring 120. However, the first write current I_{w1} does not pass through the second layer 140, which is a tunnel barrier layer.

[0077] As the first write current I_{w1} is applied to the first wiring 120, the first write current I_{w1} flows to the first layer 130 and thus a spin current is generated in the first layer 130 according to a spin Hall effect of the first layer 130, and a coercive force of the first layer 130 is less than before the first write current I_{w1} flows according to the interaction of the spin current and the magnetic moment of the first layer 130. For example, when the first write current I_{w1} is applied, the magnetic hysteresis characteristic curve of the first layer 130 has a second width $H2$ in the horizontal direction. The second width $H2$ is smaller than the first width $H1$ of FIG. 6. Accordingly, the magnetic moment of the first layer 130 is rotated by a specific or predetermined amount in a

given direction, and as a result, a plurality of data storage layers MSs of unit cells aligned on the first wiring **120** may be in a half-selected state.

[0078] Next, as shown in FIG. 7, a second write current I_{w2} is applied to the data storage layer MS of the selected unit cell in which data is to be recorded in a state in which the first write current I_{w1} is applied. As a result, the first and second write currents I_{w1} and I_{w2} may be simultaneously or concurrently applied thereto. For description, in FIG. 7, the rightmost unit cell **700** is selected as the selected unit cell.

[0079] The second write current I_{w2} applied to the unit cell **700** selected through the second wiring **170** which may be a word line flows to the first wiring **120** through the third layer **150**, the second layer **140**, and the first layer **130** in turn. In this process, the magnetic moment of the first layer **130** may be switched in the same direction or in the opposite direction to the fixed magnetic moment of the third layer **150**, which is the pinned layer, by the spin transfer torque generated in the tunnel junction.

[0080] For example, the coercive force of the first layer **130** lowered while the first write current I_{w1} is applied may disappear while the second write current I_{w2} is applied. This suggests that the magnetic moment of the first layer **130** is switched or completely switched in the same direction or the opposite direction as the fixed magnetic moment of the third layer **150** while the first and second write currents I_{w1} and I_{w2} are simultaneously applied to the data storage layer MS.

[0081] The magnetic hysteresis characteristic curve below the selected unit cell **700** indicates that the magnetic moment of the first layer **130** is completely switched in accordance with the applications of the first and second write currents I_{w1} and I_{w2} .

[0082] As described above, in a data recording process for the selected unit cell **700**, the spin current generated by the first and second write currents I_{w1} and I_{w2} is used, and a separate external magnetic field is not used. Accordingly, a problem caused by the use of an external magnetic field of the existing SOT-MRAM may be solved or reduced in likelihood of occurrence and/or in impact from occurrence.

[0083] In some examples, the operation of erasing (deleting) the recorded data may be performed opposite to the operation of writing data. For example, in FIG. 7, the erasing operation may be performed by applying an erasing current in a direction opposite to the second write current I_{w2} while the first write current I_{w1} is maintained. In this erasing operation, an application direction of the first write current I_{w1} may be the same as that of FIG. 7 or may be opposite to that of FIG. 7. In the erasing operation, the first write current I_{w1} of FIG. 7 may be represented by a first erasing current, and the erasing current applied in a direction opposite to the second write current I_{w2} may be represented by a second erasing current. In some examples, a magnitude of the first erasing current may be the same as or different from a magnitude of the first write current I_{w1} , and a magnitude of the second erasing current may be the same as or different from a magnitude of the second write current I_{w2} .

[0084] Next, an electronic apparatus according to some example embodiments will be described. The electronic apparatus according to an example embodiment may include a magnetic memory device (e.g., SOT-MRAM) according to the example embodiment described above.

[0085] FIG. 8 is a schematic block diagram of a display device **1420** including a display driver IC (DDI) **1400** and a DDI **1400** as a first electronic apparatus according to some example embodiments.

[0086] Referring to FIG. 8, the DDI **1400** may include a controller **1402**, a power supply circuit **1404**, a driver block **1406**, and a memory block **1408**. The controller **1402** may receive and decode a command applied from the main processing unit (MPU) **1422**, and may control each of blocks of the DDI **1400** to implement an operation according to the command. The power supply circuit unit **1404** generates a driving voltage in response to the control of the controller **1402**. The driver block **1406** drives the display panel **1424** using the driving voltage generated by the power supply circuit unit **1404** in response to the control of the controller **1402**. The display panel **1424** may be a liquid

crystal display panel or a plasma display panel. The memory block **1408** may include a volatile memory (e.g., RAM) and/or a nonvolatile memory as a block for temporarily storing a command input to the controller **1402** or control signals output from the controller **1402**, or a block for storing necessary data. In some examples, the nonvolatile memory may include a ROM and/or a magnetic memory device according to the above-described example embodiment.

[0087] FIG. **9** is a block diagram illustrating an electronic system **1800** as a second electronic apparatus according to an example embodiment.

[0088] Referring to FIG. **9**, the electronic system **1800** includes a memory **1810** and a memory controller **1820**. The memory controller **1820** may control the memory **1810** to read data from the memory **1810** and/or write data to the memory **1810** in response to a request from the host **1830**. In some examples, the memory **1810** may include a magnetic memory device according to various example embodiments.

[0089] FIG. **10** is a block diagram illustrating an electronic system **1900** as a third electronic apparatus according to some example embodiments.

[0090] Referring to FIG. **10**, the electronic system **1900** may configure a wireless communication device, or an apparatus capable of transmitting and/or receiving information under a wireless environment. The electronic system **1900** includes a controller **1910**, an input/output device (I/O) **1920**, a memory **1930**, and a wireless interface **1940** which are interconnected through a bus **1950**, respectively.

[0091] The controller **1910** may include at least one of a microprocessor, a digital signal processor, or a processing device similar thereto. The input/output device **1920** may include at least one of a keypad, a keyboard, and a display.

[0092] The memory **1930** may be used to store commands executed by the controller **1910**. For example, the memory **1930** may be used to store user data. In some examples, the memory **1930** may include a magnetic memory device according to various example embodiments.

[0093] The electronic system **1900** may use the wireless interface **1940** to transmit/receive data through a wireless communication network. The wireless interface **1940** may include an antenna and/or a wireless transceiver. In some example embodiments, the electronic system **1900** may be used in a third generation communication system, e.g., a communication interface protocol of the third generation communication system, such as code division multiple access (CDMA), global system for mobile communications (GSM), north American digital cellular (NADC), extended-time division multiple access (E-TDMA), and/or wideband code division multiple access (WCDMA).

[0094] FIG. **11** is a block diagram schematically illustrating a fourth electronic apparatus according to some example embodiments.

[0095] Referring to FIG. **11**, in a network environment **2200**, an electronic apparatus **2201** may communicate with another electronic apparatus **2202** through a first network **2298** (a short-range wireless communication network or the like), or with another electronic apparatus **2204** and/or a server **2208** through a second network **229** (a long-range wireless communication network or the like). The electronic apparatus **2201** may communicate with an electronic apparatus **2204** through a server **2208**. The electronic apparatus **2201** may include a processor **2220**, a memory **2230**, an input device **2250**, an acoustic output device **2255**, a display device **2260**, an audio module **2270**, a sensor module **2210**, an interface **2277**, a haptic module **2279**, a camera module **2280**, a power management module **2288**, a battery **2289**, a communication module **2290**, a subscriber identity module **2296**, and/or an antenna module **2297**. Some (such as the display device **2260**) of these components may be omitted from or other components may be added to the electronic apparatus **2201**. Some of these components may be implemented as one integrated circuit. For example, a fingerprint sensor **2211**, or an iris sensor, an illuminance sensor, etc., of the sensor module **2210** may be embedded in the display device **2260** (such as a display, etc.).

[0096] The processor **2220** may execute software (program **2240** or the like) to control one or a

plurality of other components (hardware and software components, or the like) of the electronic apparatus **2201** connected to the processor **2220**, and may perform processing or operations of various data. As part of processing or operations of data, the processor **2220** may load commands and/or data received from other components (the sensor module **2210**, the communication module **2290**, etc.) into the volatile memory **2232**, process commands and/or data stored in the volatile memory **2232**, and store the resultant data in the nonvolatile memory **2234**. The processor **2220** may include a main processor **2221** (a central processing unit, an application processor, etc.), and an auxiliary processor **2223** (graphics processing unit, image signal processor, sensor hub processor, communication processor, etc.) that may be operated independently of or together with the main processor. The auxiliary processor **2223** may use less power than the main processor **2221** and perform a specialized function.

[0097] The auxiliary processor **2223** may control functions and/or states related to some (the display device **2260**, sensor module **2210**, communication module **2290**, etc.) of the components of the electronic apparatus **2201**, in place of the main processor **2221** while the main processor **2221** is in an inactive state (sleep state), or together with the main processor **2221** while the main processor **2221** is in an active state (application execution state). The auxiliary processor **2223** (image signal processor, communication processor, etc.) may be implemented as part of other functionally related components (the camera module **2280**, communication module **2290**, etc.).

[0098] The memory **2230** may store various data required by the components of the electronic apparatus **2201** (the processor **2220**, the sensor module **2276**, etc.). The data may include, for example, software (program **2240**, etc.) and input data and/or output data for the command associated therewith. The memory **2230** may include volatile memory **2232** and/or nonvolatile memory **2234**. The nonvolatile memory **2234** may include a built-in memory **2236** and an external memory **2238**. In some examples, the nonvolatile memory **2234** may include a magnetic memory device using a spin current according to the above-described example embodiment.

[0099] The program **2240** may be stored in the memory **2230** as software, and may include an operating system **2242**, middleware **2244**, and/or an application **2246**.

[0100] The input device **2250** may receive commands and/or data to be used in components (processor **2220**) of the electronic apparatus **2201** from the outside (user, etc.) of the electronic apparatus **2201**. The input device **2250** may include a microphone, a mouse, a keyboard, and/or a digital pen (such as a stylus pen, etc.).

[0101] The sound output device **2255** may output the sound signal to the outside of the electronic apparatus **2201**. The sound output device **2255** may include a speaker and/or a receiver. Speakers may be used for general purposes such as multimedia playback or recording playback, and receivers may be used to receive incoming calls. The receiver may be coupled as part of a speaker or may be implemented as an independent separate device.

[0102] The display device **2260** may visually provide information to the outside of the electronic apparatus **2201**. The display device **2260** may include one or more of a display, a hologram device, or a projector and a control circuit for controlling the corresponding devices. The display device **2260** may include a touch circuitry configured to sense a touch, and/or a sensor circuit (a pressure sensor, etc.) configured to measure an intensity of a force generated by the touch.

[0103] The audio module **2270** may convert sound into an electrical signal or conversely convert the electrical signal into sound. The audio module **2270** may acquire sound through the input device **2250**, or output sound through a communication module, and/or a speaker and/or a headphone of another electronic apparatus (e.g., electronic apparatus **2102**) directly or wirelessly connected to the electronic apparatus **2201**.

[0104] The sensor module **2210** may detect an operating state (power, temperature, etc.) of the electronic apparatus **2201** or an external environmental state (user state, etc.) of the electronic apparatus **2201** and generate an electrical signal and/or a data value corresponding to the sensed state. The sensor module **2210** may include a fingerprint sensor **2211**, an acceleration sensor **2212**,

a position sensor **2213**, a 3D image sensor **2214**, etc. In addition, the sensor module **2210** may include one or more of an iris sensor, a gyro sensor, an atmospheric pressure sensor, a magnetic sensor, a grip sensor, a proximity sensor, a color sensor, an infrared sensor, a biometric sensor, a temperature sensor, a humidity sensor, and/or an illuminance sensor.

[0105] The 3D image sensor **2214** senses the shape and movement of an object by irradiating a predetermined light to the object and analyzing the light reflected from the object, and may include a meta-optical device.

[0106] The interface **2277** may support one or more specified protocols that can be used to connect the electronic apparatus **2201** directly or wirelessly with other electronic apparatus (electronic apparatus **2202**, etc.). The interface **2277** may include one or more of a high definition multimedia interface (HDMI), a universal serial bus (USB) interface, an SD card interface, and/or an audio interface.

[0107] The connection terminal **2278** may include a connector through which the electronic apparatus **2201** can be physically connected to another electronic apparatus (e.g., the electronic apparatus **2102**). The connection terminal **2278** may include one or more of an HDMI connector, a USB connector, an SD card connector, and/or an audio connector (such as a headphone connector).

[0108] The haptic module **2279** may convert an electrical signal into a mechanical stimulus (vibration, movement, etc.) or electrical stimulus that can be recognized by the user through touch or motor sensation. The haptic module **2279** may include one or more of a motor, a piezoelectric element, and/or an electrical stimulus device.

[0109] A camera module **2280** may capture a still image and a moving image. The camera module **2280** may include a lens assembly including one or more lenses, image sensors, image signal processors, and/or flashes. The lens assembly included in the camera module **2280** may collect light emitted from an object to be photographed.

[0110] The power management module **2288** may manage power supplied to the electronic apparatus **2201**. The power management module **388** may be implemented as part of a power management integrated circuit (PMIC).

[0111] The battery **2289** may supply power to components of the electronic apparatus **2201**. The battery **2289** may include a non-rechargeable primary battery, a rechargeable secondary battery, and/or a fuel cell.

[0112] The communication module **2290** may support establishing a direct (wired) communication channel and/or a wireless communication channel between the electronic apparatus **2201** and another electronic apparatus (electronic apparatus **2102**, electronic apparatus **2104**, server **2108**, etc.), and performing communication through the established communication channel. The communication module **2290** may include one or more communication processors that operate independently of the processor **2220** (such as an application processor) and support direct communication and/or wireless communication. The communication module **2290** may include one or more of a wireless communication module **2292** (a cellular communication module, a short-range wireless communication module, a Global Navigation Satellite System (GNSS) communication module, etc.), and/or a wired communication module **2294** (a local area network (LAN) communication module, power line communication module, etc.). A corresponding communication module of these communication modules may enable one electronic apparatus to communicate with other electronic apparatuses through a first network **2298** (a short-range communication network such as one or more of Bluetooth, Wi-Fi Direct, or infrared data association (IrDA)) or a second network **2299** (a long-range communication network such as a cellular network, Internet, or computer network (such as LAN, WAN, etc.)). These various types of communication modules may be integrated into a single component (such as a single chip), or may be implemented as a plurality of separate components (multiple chips). The wireless communication module **2292** may identify and authenticate the electronic apparatus **2201** in a communication network such as the first network **2298** and/or the second network **2299** using

subscriber information (such as an international mobile subscriber identifier (IMSI) etc.) stored in a subscriber identity module **2296**.

[0113] The antenna module **2297** may transmit a signal and/or power to the outside (such as another electronic apparatus) or receive such a signal and/or power from the outside. The antenna may include a radiator formed of a conductive pattern formed on a substrate (PCB, etc.). The antenna module **2297** may include one or a plurality of antennas. When a plurality of antennas are included in the antenna module **2297**, an antenna suitable for a communication scheme used in a communication network such as a first network **2298** and/or a second network **2299** is selected among the plurality of antennas by the communication module **2290**. A signal and/or power may be transmitted or received between the communication module **2290** and another electronic apparatus through the selected antenna. Other components (RFIC, etc.) in addition to the antenna may be included as part of the antenna module **2297**.

[0114] Some of the components are connected to each other and may exchange signals (commands, data, etc.) via a communication scheme (one or more of a bus, General Purpose Input and Output (GPIO), Serial Peripheral Interface (SPI), Mobile Industry Processor Interface (MIPI), etc.) and can interchange signals (commands, data, etc.) between peripherals.

[0115] The commands or data may be transmitted and/or received between the electronic apparatus **2201** and the external electronic apparatus **2204** through the server **2108** connected to the second network **2299**. The other electronic apparatuses **2202** and **2204** may be the same as or different types of apparatuses from the electronic apparatus **2201**. All or some of the operations executed in the electronic apparatus **2201** may be executed in one or more of the other electronic apparatuses **2202** and **2204**, and the server **2208**. For example, when an electronic apparatus **2201** needs to perform a function or service, instead of executing the function or service itself, the electronic apparatus **2201** may request one or more other electronic apparatus to perform part or all of the function or service. One or more other electronic apparatuses receiving the request may execute an additional function or service related to the request and transmit a result of the execution to the electronic apparatus **2201**. To this end, cloud computing, distributed computing, and/or client-server computing technology may be used.

[0116] At intersections between the first wiring **120** and the second wiring **170**, there may be a data storage layer.

[0117] The bit line of the SOT-MRAM is a wiring of a metal material having low electrical resistance equal to or similar to a metal wiring used in a semiconductor device or an electronic apparatus, while having no spin Hall effect or substantially no spin Hall effect. Thus, the operating current density of the SOT-MRAM disclosed may be lower than that of the conventional SOT-MRAM using, as a bit line, a SOT layer (e.g., W, Ta, or Pt) having a relatively high electrical resistance as compared to the metal material. For example, in the case of the disclosed SOT-MRAM, the operating current density may be reduced to 1 MA/cm² that is about 1/100 to 1/10 compared to the operating current density of the conventional SOT-MRAM. Therefore, the number of MTJ cells that can be connected per bit line may be increased compared to the existing SOT-MRAM, which suggests that the degree of integration may be higher than that of the existing SOT-MRAM when the disclosed SOT-MRAM is used.

[0118] Alternatively or additionally, the SOT-MRAM is configured to form the free layer of the MTJ cell into a two-dimensional van der Waals material with ferromagnetic properties and metal properties while having a spin Hall effect. Accordingly, when a current flows in the free layer of the MTJ cell, a spin current is generated in the free layer, and thus a magnetic moment of the free layer may be rotated by a predetermined angle in a given direction, thereby lowering a coercive force of the free layer. These results are the same as the operating state of the MTJ cell of the existing SOT-MRAM, which suggests that a combination of the bit line of metal wiring with no spin Hall effect and the two-dimensional van der Waals material layer (free layer) with magnetic and metal properties as the disclosed SOT-MRAM has increased integration and decreased operation current

density, while having the operation characteristics (such as the write operation) of the existing SOT-MRAM.

[0119] Alternatively or additionally, the operation of the disclosed SOT-MRAM reduces the coercive force of the free layer with a first write current flowing in the free layer without passing through the tunnel barrier layer, and then fully switches the magnetic moment of the free layer with a second write current passing through the pinned layer, the tunnel barrier layer, and the free layer of the MTJ cell in turn. For example, a spin current is generated in the free layer as the second write current passes through the free layer, and the magnetic moment of the free layer is completely switched by interaction between the magnetic moment of the free layer and the generated spin current.

[0120] As described above, in the case of the disclosed SOT-MRAM, a spin current is used for the data recording operation, and an external magnetic field applied from the outside of the MTJ cell is not used. Accordingly, when the disclosed SOT-MRAM is used, a problem caused by the use of an external magnetic field in an operation of the existing SOT-MRAM may be solved, or reduced in likelihood of occurrence and/or in impact from occurrence. Alternatively or additionally, since the second write current may be less than the first write current in the data recording operation, it may be seen that the data write path and the data read path are substantially separated in the disclosed SOT-MRAM operation.

[0121] Therefore, the disclosed SOT-MRAM may maintain advantages of the existing SOT-MRAM, for example, high-speed switching characteristics of 1 nanosecond or less (<1 nsec) and high endurance.

[0122] Any of the elements and/or functional blocks disclosed above may include or be implemented in processing circuitry such as hardware including logic circuits; a hardware/software combination such as a processor executing software; or a combination thereof. For example, the processing circuitry more specifically may include, but is not limited to, a central processing unit (CPU), an arithmetic logic unit (ALU), a digital signal processor, a microcomputer, a field programmable gate array (FPGA), a System-on-Chip (SoC), a programmable logic unit, a microprocessor, application-specific integrated circuit (ASIC), etc. The processing circuitry may include electrical components such as at least one of transistors, resistors, capacitors, etc. The processing circuitry may include electrical components such as logic gates including at least one of AND gates, OR gates, NAND gates, NOT gates, etc.

[0123] In certain block diagrams it is to be understood that, unless otherwise stated or clear from context, any component may be connected to any other component, capable of one way communication and/or two-way communication and/or broadcast communication, to exchange information such as but not limited to data and/or commands. The connection may be wireless, and/or may be wired. The information may be exchanged in a serial manner, and/or in a parallel manner. The information may be or may include digital information, and/or may be or may include analog information.

[0124] It should be understood that embodiments described herein should be considered in a descriptive sense only and not for purposes of limitation. Descriptions of features or aspects within each various example embodiment should typically be considered as available for other similar features or aspects in other example embodiments. While one or more example embodiments have been described with reference to the figures, it will be understood by those of ordinary skill in the art that various changes in form and details may be made therein without departing from the spirit and scope as defined by the following claims.

Claims

1. A magnetic memory comprising: a first wiring; a second wiring spaced apart from the first wiring and intersecting the first wiring; and a data storage layer between the first wiring and the

second wiring and in contact with the first and second wirings, wherein the data storage layer comprises: a pinned layer having a fixed magnetic moment; a free layer spaced apart from the pinned layer and having a switchable magnetic moment; and an insulating tunnel barrier layer between the pinned layer and the free layer, wherein the first wiring is a wiring configured to supply a first current to generate a first spin current in the free layer, and wherein the second wiring is a wiring configured to supply a second current to generate a second spin current in the free layer.

2. The magnetic memory of claim 1, wherein a wiring contacting the free layer from among the first and second wirings includes a conductive wiring that does not have a spin Hall effect, and the free layer comprises a two-dimensional material having a spin Hall effect, magnetic properties, and metallic properties.
3. The magnetic memory of claim 2, wherein the two-dimensional material comprises a two-dimensional van der Waals material.
4. The magnetic memory of claim 3, wherein the two-dimensional van der Waals material comprises at least one of $\text{Fe.sub.(x)GeTe.sub.2}$ (x is 3 to 5), CrTe.sub.2 , and VSe.sub.2 .
5. The magnetic memory of claim 1, wherein the first wiring includes a plurality of first wirings and the second wiring includes a plurality of second wirings, and the data storage layer is provided for each intersecting area of the plurality of first wirings and the plurality of second wirings.
6. The magnetic memory of claim 3, wherein the conductive wiring comprises a Cu wiring.
7. The magnetic memory of claim 1, wherein the tunnel barrier layer comprises one or more of an MgO layer, a boron nitride (BN) layer, or an aluminum oxide layer.
8. The magnetic memory of claim 1, wherein a material of the pinned layer is same as a material of the free layer.
9. The magnetic memory of claim 1, wherein at a cross-point a width of the second wiring along a first direction is less than a width of the data storage layer along the first direction.
10. An operating method of a magnetic memory including a magnetic tunnel junction (MTJ) including a pinned layer and a free layer, the method comprising: applying a first current to the MTJ; applying a second current to the MTJ; wherein one of the first current and the second current passes through the MTJ and the other of the first current and the second current passes through only the free layer, and wherein a first spin current is generated in the free layer by the first current and a second spin current is generated in the free layer by the second current.
11. The operating method of claim 10, wherein the first and second spin currents correspond to currents for data recording.
12. The operating method of claim 10, wherein the first and second spin currents correspond to currents for data erasure.
13. The operating method of claim 10, wherein the first current is applied to a wiring contacting the free layer to lower a coercive force of the free layer.
14. The operating method of claim 13, wherein the second current is applied to a wiring contacting the pinned layer to switch a magnetic moment of the free layer in which the coercive force is lowered.
15. The operating method of claim 14, wherein the second current is less than the first current.
16. The operating method of claim 10, wherein the free layer comprises a two-dimensional van der Waals material which at room temperature has a spin Hall effect, magnetic properties, and metallic properties.
17. The operating method of claim 16, wherein, the two-dimensional van der Waals material comprises at least one of $\text{Fe.sub.(x)GeTe.sub.2}$ (x is 3 to 5), CrTe.sub.2 , and VSe.sub.2 .
18. The operating method of claim 13, wherein the wiring contacting the free layer comprises a metal wiring having no spin Hall effect.
19. The operating method of claim 10, wherein the magnetic memory comprises a spin-orbit torque (SOT)-based magnetic memory.

20. An electronic apparatus comprising a memory, wherein the memory comprises the magnetic memory of claim 1.
